



Preface

**Carlos Areces¹ · Diana Costa² · Nina Gierasimczuk³ ·
Fernando R. Velázquez-Quesada⁴**

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Building on the ideas of Floyd-Hoare logic, Dynamic Logic (DL) was introduced in the 1970s as a formal tool for reasoning about and verifying classic imperative programs. Conceptually, its main feature is that it allows us to reason not only about the states in which a program might be, but also about the structured actions that define transitions between those states. Technically, its most distinguishing feature relative to other modal logics is that it gives actions a structure. Rather than being limited to abstract 'atomic' steps, actions can be built through sequential composition ($\alpha; \beta$), non-deterministic choice ($\alpha \cup \beta$), or the finite repetition (α^*) of other actions. This internal structuring allows the logic to treat programs as first-class citizens within the formula itself.

Over time, the aim of DL has evolved and expanded beyond its original scope. It is now viewed as a general set of ideas, tools, and techniques devised for representing, describing, and reasoning about diverse types of actions. This evolution includes frameworks tailored for specific programming paradigms (e.g., separation logics), as well as settings designed for modelling new computing domains (e.g., probabilistic, continuous and quantum computation). Furthermore, the reach of DL extends to frameworks for reasoning about information dynamics (e.g., dynamic epistemic log-

✉ Carlos Areces
carlos.areces@unc.edu.ar

Diana Costa
dfdcosta@ciencias.ulisboa.pt

Nina Gierasimczuk
nigi@dtu.dk

Fernando R. Velázquez-Quesada
Fernando.VelazquezQuesada@uib.no

¹ Facultad de Matemática, Astronomía, Física y Computación, Universidad Nacional de Córdoba, Haya de la Torre s/n, Ciudad Universitaria, Córdoba 5000, Argentina

² LASIGE, Faculty of Sciences of the University of Lisbon, Campo Grande, Lisbon 1749 - 016, Portugal

³ Department of Mathematics and Computer Science, Technical University of Denmark, Richard Petersens Plads, Lyngby 2800, Denmark

⁴ Department of Information Science and Media Studies, University of Bergen, Fosswinckels gate 6, Bergen 5007, Norway

ics) and systems for reasoning about long-term information dynamics (e.g., learning theory).

Both its theoretical relevance and practical potential make DLs a topic of interest in various scientific venues, from wide-scope software engineering conferences to events specific to modal logics. One of such venues is the *Dynamic Logic—New Trends and Applications* (DaLí) Workshop. Its goal has been to bring together researchers who share interests in the formal study of actions, to create a platform for presenting ongoing research, to foster discussions and encourage new collaborations. The DaLí Workshop has taken place in Brasília (2017), Porto (2019), online (2020, 2022), Tbilisi (2023) and Xi'an (2025). The proceedings of the first five editions (respectively, Madeira and Benevides (2018); Barbosa and Baltag (2020); Martins and Sedlár (2020); Areces and Costa (2023); Gierasimczuk and Velázquez-Quesada (2024)) have collected a number of articles discussing different DLs aspects. Additionally, there have been Special Issues with revised and extended versions of papers accepted in the first three editions (Benevides & Madeira, 2020), (Baltag & Barbosa, 2021), (Martins & Sedlár, 2023).

This special issue contains extended versions of selected papers presented at the 4th and 5th editions of the Workshop, accompanied by other papers on the topic of DL obtained through an open call. The thirteen articles accepted to the Special Issue can be categorized into four distinct groups based on their primary focus: the dynamic aspects of the cognitive states of agents, the logic of actions and plans, the structural evolution of systems, and the foundational semantic architecture of logical frameworks.

Multi-Agent Epistemic Dynamics and Social Reasoning

1. John Ioki Lindqvist, Fernando Raymundo Velázquez-Quesada, and Thomas Ågotnes, *Variations on distributed belief*.
2. Jakob Dirk Top, Rineke Verbrugge, and Harmen de Weerd, *A Dynamic Epistemic Logic with Lying and Theory of Mind Limitations*.
3. Thomas Bolander and Hermine J. Grosinger, *Epistemic Proactivity via Reasoning about Beliefs and Expectations using Plausibility Models*.
4. Alessandro Aldini and Ludovico Fusco, *Bridging concurrency theory and epistemic models: a formal framework for dynamic multi-agent systems*.

These articles center on the formalization of agents' cognitive states: knowledge, belief, and expectations; and on how these states are distributed, limited, or updated within a group. They treat the agent's internal cognitive perspective as the primary object of study. For example, Lindqvist et al. introduce 'cautious' and 'bold' distributed belief to ensure group reasoning does not collapse when agents hold conflicting information. Top et al. model 'Theory of Mind' orders to represent how agents recursively attribute beliefs to others and process deception. Bolander & Grosinger employ plausibility models to distinguish between *a priori* expectations about future outcomes and *a posteriori* knowledge gained after an action. Aldini & Fusco bridge concurrency theory with epistemic logic to create a model for verifying dynamic multi-agent systems where agents possess varying observational powers.

Logics of Action, Strategic Interaction, and Causality

1. Carlos Areces, Valentin Cassano, Pablo Castro, and Raul Fervari, *Deontic Action Logics: A Modular Algebraic Perspective*.

2. Andreas Herzig and Abdallah Saffidine, *A Note on Halpern's Theory of Causality: Structural Equations and Interventions as Dynamic Logic Programs*.
3. Carlos Areces, Raul Fervari, Andrés R. Saravia, and Fernando R. Velázquez-Quesada, *The Dynamics of Knowing How*.
4. Qian Chen and Dazhu Li, *On the Logic of the Hide and Seek Game*.

These articles focus on the interaction between an agent and the environment, specifically through norms, causal interventions, executable plans, and game-theoretic rules. They move beyond static beliefs to analyze the *execution and consequences* of actions. Areces et al. provide an algebraic perspective on Deontic Action Logic (DAL), defining normative concepts like permission and prohibition as algebraic mappings. Herzig and Saffidine embed causality theory into dynamic logic, associating interventions and structural equations with logic programs to reduce causal reasoning to a model-checking problem. Areces et al. explore the 'knowing how' operator, which evaluates an agent's ability to achieve a goal based on the indistinguishability of available plans. Chen & Li investigate the logic of the 'Hide and Seek' game on graphs.

Structural Updates and Spatio-Temporal Systems

1. Penghao Du and Qian Chen, *Hilbert-Style Calculus and Tableau Calculus for Logics of Link Variations*.
2. Alexandru Baltag, Johan van Benthem, and Dazhu Li, *Temporal Dependencies in Dynamical Systems: A Modal-Style Exploration*.
3. Laura Bussi, Vincenzo Ciancia, Fabio Gadducci, and Mieke Massink, *Spatio-temporal Model Checking with VoxLogicA*.

This group is focused on the dynamic evolution of the system's underlying structure, specifically regarding spatial relations, temporal traces, and graph connectivity. The contributions prioritize the mathematical evolution of the environment or graph structure over time or space. Du & Chen examine 'link variations', providing calculi for updates that cut, add, or rotate links within accessibility relations. Baltag et al. explore functional dependencies in discrete dynamical systems, where variable values evolve through transition functions mapping states to next states. Bussi et al. describe the VoxLogicA model checker, which performs spatial image analysis over temporal frames to track evolving properties, such as movements in a video.

Syntactic and Semantic Model Foundations

1. Kai F. Wehmeier, *Binding in classical and dynamic predicate logic*.
2. Manuel A. Martins, Abílio Rodrigues, Marcelo Coniglio, and Alfredo Freire, *On many-logic modal structures and information-based logics*.

These articles address the foundational meta-logical properties of formal languages, such as the mechanisms of variable binding and the integration of multiple underlying logics. They deal with the fundamental architecture of logical systems rather than specific application domains like games or robotics. Wehmeier provides a deep investigation into variable binding in dynamic and classical predicate logic, establishing a single unifying binding schema. Martins et al. introduce 'many-logic modal structures' (MLMS), which allow different worlds within a single Kripke model to

be governed by different underlying logics (such as paraconsistent, paracomplete, or classical systems) by anchoring them to a base lattice.

It is a pleasure to present this Special Issue to the research community. The papers collected here continue a rich tradition of inquiry while at the same time attesting to the ongoing vitality of the field. The contributions also reflect the field's strongly interdisciplinary character, engaging with and drawing inspiration from cognitive science, artificial intelligence, linguistics, and related areas. In this respect, the *Journal of Logic, Language and Information* is a particularly fitting venue, given its longstanding commitment to the intersection of these disciplines. We hope that this volume will be of interest both to specialists and to researchers working in neighboring fields, and that it will contribute to further dialogue across disciplinary boundaries.

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